



Synthesis and marine antifouling properties of carbazole amide derivatives with fluorescent properties and their zinc acrylate resins

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ARTICLE INFO

Keywords:

Fluorescent antifouling
Carbazole derivatives
Zinc acrylate resin
Algae inhibition
Photosynthetic pigment

ABSTRACT

Fluorescent antifouling is a method that utilizes substances fluorescent repellent effect to prevent organisms attachment and fouling. It is considered one of the directions for developing environmentally friendly marine antifouling materials. The current fluorescent antifouling materials were characterized by issues such as single functionality, unclear antifouling mechanisms, and suboptimal antifouling performance. In this study, three carbazole derivatives containing carbon-carbon double bonds were synthesized by Friedel-Crafts alkylation reaction, and a series of zinc acrylate resins containing carbazole derivative structures were further synthesized through free radical polymerization. Three carbazole derivatives were tested for their fluorescence and their growth inhibition properties against bacteria and algae. The results showed that the three carbazole derivatives not only had high fluorescence intensity, but also had good inhibitory effects on bacteria and algae, with the highest inhibition rate of 81% on algae. The fluorescence properties, algal inhibition properties and anti-algal adhesion properties of the resin were investigated. The results showed that the fluorescence intensity of the resin was enhanced with the increase of carbazole derivative content. The inhibition rate of the resin against algae was as high as 88.5%, and the anti-algal adhesion rate of the resin was improved under light conditions. Mechanistic studies showed that the resin influenced the cell wall, and the fluorescence caused the algae to prioritize photosynthesis rather than attachment and settlement. Self-polishing performance test results showed that the introduction of carbazole derivatives improved the stability of the resin. The results of the real sea antifouling test show that the resin integrates fluorescence, chemical activity and self-polishing, and has excellent antifouling performance. This provides a theoretical basis for the development of environmentally friendly marine antifouling materials.

1. Introduction

Marine fouling refers to the harmful accumulation of marine organisms on ships and marine facilities, which is a major challenge for humans in the process of developing and utilizing the ocean [1,2]. It can lead to increased fuel consumption of ships, blockage of submarine pipelines, damage to marine instrumentation facilities, etc. [3,4]. Applying marine antifouling coatings has been considered the most economical and effective method to solve the problem of marine biofouling [5]. Although organotin antifouling coatings had possessed high-efficiency antifouling properties, their use was prohibited due to

significant toxicity to marine organisms and the resulting damage to the marine ecological environment [6]. Based on the antifouling mechanism of organic tin antifouling coatings, tin-free self-polishing antifouling coatings have been developed. The widely used tin-free self-polishing antifouling coatings rely on high cuprous oxide content to achieve efficient antifouling performance. However, studies have shown that the extensive use of cuprous oxide has caused potential harm to the marine environment. Therefore, the functionality of resins in antifouling coatings needed to be enhanced [7].

The functions of resin in antifouling coatings mainly include self-polishing, low surface energy, and so on [8–11]. However, the

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<https://doi.org/10.1016/j.cej.2024.158017>

Received 1 September 2024; Received in revised form 30 October 2024; Accepted 23 November 2024

Available online 25 November 2024

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antifouling performance of antifouling coatings based on a single functional resin is still not ideal. Therefore, improving the antifouling performance through the multi-functional synergy of the resin is a hot spot in the current research and development in this field.

The fluorescent antifouling was a method that utilized the fluorescent repellent effect of substances to prevent the attachment and fouling of organisms. Research has shown that fluorescence could alter the chloroplast morphology of diatoms and effectively inhibit the accumulation and settlement on coating surfaces, thus contributing to the early prevention and control of marine biofouling [12–14]. Additionally, in transparent seawater, it was noted that ultraviolet rays could penetrate up to 20 m, making fluorescent antifouling feasible [15].

For example, Cao et al. prepared an antifouling composite coating of colored phosphor, and the results of algae inhibition showed that the antifouling composite coating of colored phosphor could effectively reduce the formation of biofilm and the attachment of diatoms [14]. Xiong et al. added two kinds of long afterglow phosphors with fluorescence intensity into the silicone coatings to prepare composite antifouling coatings with different colors and brightness, and the results of algae inhibition experiments showed that the higher the fluorescence intensity, the better the antifouling performance of the coatings [16]. Inspired by the antifouling ability of fluorescent coral, Jin et al. prepared a coating with a sandwich structure containing fluorescent powder, and the results of bacterial inhibition experiments showed that the coating had a good bacterial inhibition effect [17]. Hao et al. introduced long afterglow phosphors into silicone elastomers by physical blending to prepare a composite coating, which exhibited excellent antifouling performance due to the synergistic effect of slow release and fluorescent antifouling [18]. However, long-term experimental results have indicated that the method involving the addition of fluorescent powders for antifouling effects was found to be unstable. Consequently, it has been recognized that incorporating fluorescent groups into the resin structure and harnessing the multifunctional synergy of the resin are effective approaches to improve the antifouling performance of antifouling coatings.

Carbazole compounds, known for their strong fluorescence, have been extensively utilized in the biopharmaceutical field, being employed as anti-bacterial agents and in the production of anti-cancer drugs and antibiotics [19,20]. Capsaicin derivatives exhibit good antibacterial, algacidal, and antifouling properties under their hydroxyl, amide, and other active groups [21,22]. Integrating the fluorescence of carbazoles and the chemical activity of capsaicin derivatives to prepare multifunctional marine antifouling materials from them is a new development direction. In this paper, three carbazole amide derivatives containing carbon–carbon double bonds were synthesized by Friedel-Crafts alkylation reaction, and zinc acrylate resins containing carbazole derived structures were prepared with them as functional monomers. The antifouling properties of the resin were also investigated, providing a theoretical basis for developing environmentally friendly marine antifouling materials.

2. Experimental

2.1. Materials

Carbazole, N-phenylcarbazole, N-ethylcarbazole, N-Methylolacrylamide, acrylic monomers, and benzoic acid were purchased from Shanghai Aladdin Biochemical Technology Co. Anhydrous ethanol, concentrated sulfuric acid, acetone, azodiisobutyronitrile were purchased from Sinopharm Chemical Reagent Co. All reagents were used according to the instructions.

2.2. Synthesis of carbazole derivatives

A three-necked flask equipped with a stirrer was placed in an oil bath and anhydrous ethanol was added as solvent. Carbazole (0.1 mol) and N-

hydroxy meth acrylamide (0.12 mol) were dispersed sequentially in a certain amount of solvent using an electric stirrer. After stirring for 1 h, concentrated sulfuric acid (10 mL) was added dropwise to a three-necked flask. The reaction was carried out at 40 °C for 72 h until a solid product was obtained. The precipitated product was filtered and the filtered solid was washed with deionized water to a pH of 7. The crude product was purified by recrystallization with anhydrous ethanol to give N-(carbazole-3-methyl) acrylamide (CZAM). The structural formula of the carbazole derivatives is shown in Fig. 1. IR (KBr) ν : C-H (Bending vibration of benzene ring): 750.62, C-N:1101.01, C = C (benzene skeleton vibration): 1492.70, 1546.83, 1629.59, C = O (Secondary amide I band): 1655.08, N-H: 3278.82 cm^{-1} ; ^1H NMR (DMSO- d_6 , 600 MHz) δ : 4.50 (d, 2H, -CH $_2$ -), 5.62 (m, 1H, -CH =), 5.81 (d, 2H, =CH $_2$), 7.22 (m, 1H, Ph-H), 7.43 (m, 2H, Ph-H), 7.74 (d, 1H, Ph-H), 7.78 (d, 1H, Ph-H), 8.02 (d, 1H, Ph-H), 8.11 (d, 1H, Ph-H), 8.61 (t, 1H, N-H), 9.12 (t, 1H, N-H); ^{13}C NMR (DMSO- d_6 , 150 MHz) δ : 43.1, 110.2, 110.6, 119.7, 119.9, 120.2, 121.1, 122.7, 123.2, 125.7, 126.6, 127.3, 131.5, 132.3, 140.2, 165.5; HRMS, m/z : 273.1001 [M + Na] $^+$.

N,N'-((2-(9-carbazolyl))-1,3-phenyl)bis(methylene)bisacrylamide (PCAM) was synthesized with reference to CZAM. IR (KBr) ν : C-H (Bending vibration of benzene ring): 801.90, C-N: 1075.37, C = C (benzene skeleton vibration): 1504.10, 1536.86, 1620.89, C = O (Secondary amide I band): 1655.08, N-H: 3291.64 cm^{-1} ; ^1H NMR (DMSO- d_6 , 600 MHz) δ : 4.53 (d, 4H, -CH $_2$ -), 5.63 (m, 2H, -CH =), 6.15 (m, 2H, =CH $_2$), 6.31 (m, 2H, =CH $_2$), 7.35 (m, 4H, Ph-H), 7.53 (t, 1H, Ph-H), 7.59 (m, 2H, Ph-H), 7.68 (t, 2H, Ph-H), 8.10 (d, 2H, Ph-H), 8.66 (t, 2H, N-H); ^{13}C NMR (DMSO- d_6 , 150 MHz) δ : 42.9, 110.1, 119.8, 123.0, 125.8, 126.8, 127.0, 131.6, 132.3, 140.1, 165.0; HRMS, m/z : 432.1696 [M + Na] $^+$.

N-((9-ethylcarbazole)-3-methyl) acrylamide (ECAM) was synthesized with reference to CZAM. IR (KBr) ν : C-H (Bending vibration of benzene ring): 804.75, C-N: 1073.95, C = C (Benzene skeleton vibration): 1492.70, 1536.86, 1622.32, C = O (Secondary amide I band): 1655.08, N-H: 3281.67 cm^{-1} ; ^1H NMR (DMSO- d_6 , 600 MHz) δ : 1.29 (d, 3H, -CH $_3$), 4.43 (q, 2H, -CH $_2$ -), 4.51 (d, 2H, -CH $_2$ -), 5.61 (m, 1H, -CH =), 6.15 (m, 1H, =CH $_2$), 6.31 (m, 1H, =CH $_2$), 7.18 (t, 1H, Ph-H), 7.42 (m, 2H, Ph-H), 7.58 (m, 2H, Ph-H), 8.04 (d, 1H, Ph-H), 8.12 (d, 1H, Ph-H), 8.66 (t, 1H, N-H). ^{13}C NMR (DMSO- d_6 , 150 MHz) δ : 14.1, 37.4, 43.1, 109.5, 109.6, 119.1, 119.9, 120.7, 122.5, 126.2, 126.3, 130.1, 132.4, 164.9; HRMS, m/z : 301.1319 [M + Na] $^+$. (The characterization of carbazole derivatives is shown in the Supplementary Materials Fig. S1–S4).

2.3. Fluorescence intensity test of carbazole derivatives

An amount of carbazole derivatives were weighed and dissolved in DMSO and diluted to 1×10^{-5} mol/L respectively, and the fluorescence emission spectra were measured using a fluorescence spectrophotometer. The slit width EX/EM = 2.5 nm and scanning voltage = 700 V.

2.4. Antibacterial test of carbazole derivatives

Marine Pseudoalteromonas was selected as the test strain. The marine Pseudoalteromonas was placed in a thermostatic incubator at 37 °C for 24 h. The activated bacteria were inoculated into 2216 E liquid medium using an inoculation ring and then cultured in a thermostatic incubator for 24 h. The bacterial suspension was diluted to a suitable concentration by ultraviolet–visible spectrophotometer, and then a certain amount of compound solution was added to the bacterial suspension and placed in a constant temperature oscillation incubator for 24 h. The cultured bacterial suspension was diluted 10 times with normal saline (9 g/L), and then 0.2 mL bacterial solution was absorbed on the 2216 E solid medium. The bacterial solution was uniformly coated on the surface of the solid medium using a triangular coating rod until it was completely absorbed, and then the solid medium was placed in a 37 °C constant temperature incubator for 24 h.

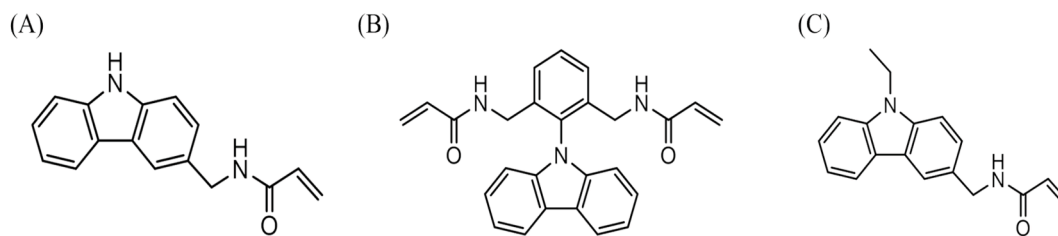


Fig. 1. Structural formula of carbazole derivatives: A. CZAM B. PCAM C. ECAM.

2.5. Algae inhibition test of carbazole derivatives

Phaeodactylum tricornutum and *Nitzschia closterium* were selected as the tested algae species. The maximum UV absorption wavelength of the two algae was 680 nm [23]. Firstly, the algae species in the exponential growth phase were diluted in turn, and the absorbance was measured at the maximum absorption wavelength. The concentration of algae was determined by blood cell counting plate. Standard work curves were obtained with absorbance as the abscissa and concentration as the ordinate.

The algal solution cultured to the exponential growth period was diluted to a suitable concentration, 100 ml of diluted algal solution was measured in a conical flask, and 1 ml (1.5 mg/ml) of carbazole derivative solution was pipetted and added to the conical flask, and mixed homogeneously. The algal solution without the test compound was used as a blank control, and the absorbance value of each algal solution was measured at the maximum absorption wavelength every 24 h. The time-algal concentration curve was plotted according to the linear relationship of algal absorbance value concentration [24]. The light intensity in the incubator was 4000 lx, the ratio of light to dark was 14:10, and the temperature was 21 °C. The seawater used was filtered through a sand core funnel and sterilized in an autoclave (121 °C). The algal inhibition rate was calculated according to the following equation:

$$I = (I_0 - I_a) / I_0 \times 100\%$$

Where I is the algal inhibition rate, I_0 is the absorbance value of the control group and I_a is the absorbance value of the experimental group.

2.6. Synthesis of zinc acrylate resins containing carbazole derivative structures

A three-necked flask was placed in an oil bath, and the solvents xylene and n-butanol (xylene: n-butanol = 4:1, wt.%) were added to raise the temperature to 85 °C, and then acrylic monomers, carbazole derivatives, and 2,2'-Azobis (2-methyl propionitrile) were added at intervals of 15 min (prepolymer 1.2 %, W%). The reaction system was then held at 85 °C for 4 h to obtain the prepolymer, which was kept at reflux throughout the process.

Finally, zinc oxide, benzoic acid, and the solvents xylene, n-butanol, and butyl acetate were added to a three-necked flask and reacted at 85 °C for 4 h. Zinc acrylate resins containing carbazole derived structures were obtained and characterized by infrared spectroscopy. (More details are shown in the [supplementary data in Table S1](#)).

As an example of the synthesis of resin PCZ, the synthetic route is shown in Fig. 2.

2.7. Fluorescence intensity testing of zinc acrylate resins containing carbazole derivative structures

The resin was dried and the fluorescence emission spectra were measured using a fluorescence spectrophotometer. The slit width EX/EM = 2.5 nm and the scanning voltage = 700 V.

2.8. Algae inhibition test of zinc acrylate resins containing carbazole derivative structures

Phaeodactylum tricornutum and *Nitzschia closterium* were selected as the tested algae species. Firstly, the samples to be tested were sterilized

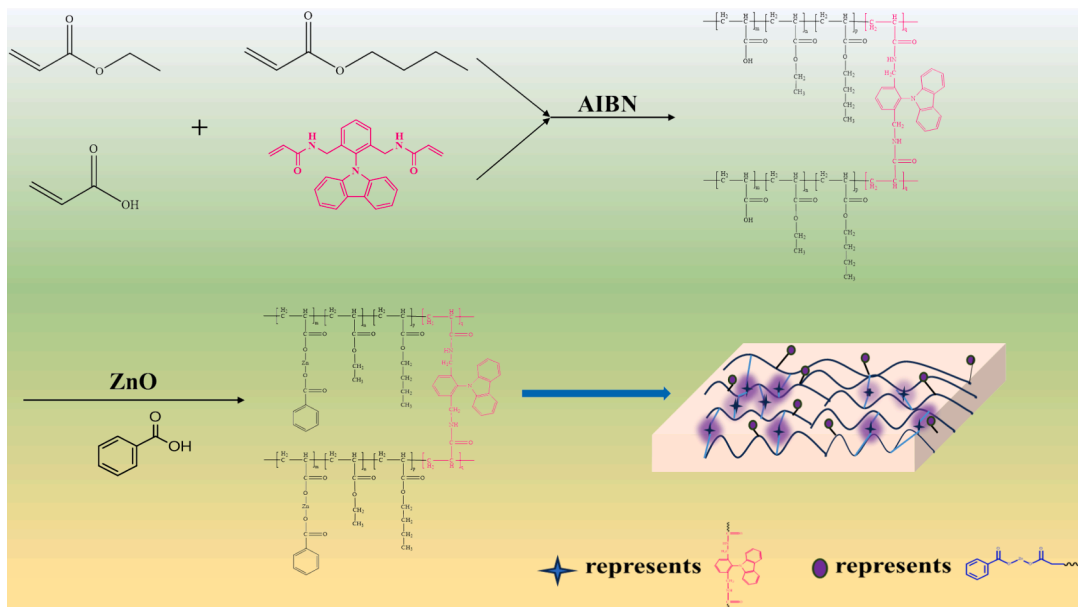


Fig. 2. Schematic representation of the synthesis of zinc acrylate resins containing carbazole derivative structures.

by UV for 24 h. Then the samples and the algae solution (100 ml) in the exponential growth period were added to Petri dishes and incubated in an artificial climate incubator and under natural light conditions using ordinary glass plates as blanks and ordinary zinc acrylate resin as control. After seven days of incubation, the absorbance of the algal solution was measured using a UV-visible spectrophotometer, and the algal inhibition rate was calculated [25,26].

2.9. Algae adhesion resistance test of zinc acrylate resins containing carbazole derivative structures

Firstly, *Phaeodactylum tricornutum* and *Nitzschia closterium* cultured to exponential growth cycle were diluted into a certain concentration of algae solution. The resin coating sample was placed at the bottom of the culture dish, and 40 mL of algae solution was added. The petri dishes were then placed in an incubator and cultured in an artificial climate incubator and under alternating natural light/dark conditions, respectively, and the algal solution was shaken at the same time every day. After 7 days, the coated samples were removed and their surfaces were rinsed with algal solution in a Petri dish to remove the weaker microalgae attached to the surface. Then all the algae attached to the coated surface were rinsed into a beaker with 200 mL of nutrient solution, and chlorophyll (a) was detected using a portable fluorometer with ordinary zinc acrylate resin as a control. The measured values can reflect the concentration of algal cells and thus the amount of algal cells attached to the resin surface.

2.10. Algae adhesion resistance test of zinc acrylate resins containing carbazole derivative structures

Photosynthetic pigments are important indicators of algal photosynthesis and play an important role in algal growth and development [27]. *Phaeodactylum tricornutum* was cultured for seven days according to the experimental method of 2.6 in this paper, and then 10 ml of algae liquid was taken and filtered by 0.7 μm glass fiber filter membrane. The filter membrane was folded and soaked in 5 mL of acetone (90 %, wt.%), shaken evenly, and then extracted for 24 h under dark conditions at 4 °C. The algal solution was centrifuged at 4500 r/min for 10 min, and then the supernatant was aspirated, and the absorbance values of the supernatant were determined using a UV-visible spectrophotometer at 440, 630, 644, 647, 662, 664, and 750 nm, with ordinary zinc acrylate resin as a control. The contents of chlorophyll (a), chlorophyll (c), and carotenoids (mg/L) were calculated according to the following formula:

$$\text{chlorophyll}(a) = [11.85(OD_{664} - OD_{750}) - 1.54(OD_{647} - OD_{750}) - 0.08(OD_{630} - OD_{750})] \times v / (V \times L)$$

$$\text{chlorophyll}(c) = [24.52(OD_{630} - OD_{750}) - 1.67(OD_{664} - OD_{750}) - 7.60(OD_{647} - OD_{750})] \times v / (V \times L)$$

$$\text{carotenoids} = (4.70OD_{440} - 1.38OD_{662} - 5.48OD_{644}) \times v / (V \times L)$$

In the formula: v (mL) and V (mL) represent the volume of pigment extract and algae liquid respectively, and L (cm) represents the optical distance of the cuvette.

2.11. Testing the effect of zinc acrylate resins containing carbazole derivative structures on the morphology of diatom cells

Referring to the experimental method in 2.6 of this paper, the *Phaeodactylum tricornutum* was cultivated for 7 days, and then an appropriate amount of the algal solution was taken to a centrifuge tube. Algal cell precipitates were collected after centrifugation using a centrifuge, and algal cells were fixed using glutaraldehyde solution at 4 °C. An appropriate amount of phosphate buffer solution (PBS) was taken and washed three times, followed by gradient dehydration of the

algal cells using 30 %, 50 %, 70 %, 80 %, 90 %, and 100 % ethanol solutions in that order. After dehydration was completed, the algal cells were freeze-dried, and then the morphology of the algal cells was observed using SEM [28].

2.12. Testing of self-polishing properties of zinc acrylate resins containing carbazole derivative structures

Zinc acrylate resin is a kind of paint-based material with a self-polishing effect. An accurate understanding of the self-polishing rate of the resin is very important for judging the service life of the resin [29]. The self-polishing properties of the coatings were tested using an in-laboratory dynamic acceleration device. Firstly, the resin was coated on the ABS plate (350 mm $\times$ 100 mm $\times$ 3 mm) and dried, and then the coated sample was placed in a dynamic acceleration experimental device. (The device diagram is shown in the [Supplementary Data](#) in Fig. S5). Every seven days, the ABS plates were washed with deionized water and dried to constant weight (37 °C) at the end of the cyclic test. The surface morphology of the coatings before and after each cycle was observed using a laser scanning confocal microscope, and the samples were weighed using an analytical balance.

2.13. Marine antifouling performance testing of zinc acrylate resins containing carbazole derivative structures

Compared with the complicated marine environment, only a few marine species are used to assess the antifouling property under experimental conditions, so an additional antifouling test in a marine environment is very necessary [30]. The antifouling performance testing in a real sea environment was conducted according to GB/T 5370-85 [31]. First, the PVC board was polished with 80 mesh sandpaper, then rinsed with anhydrous ethanol and dried. Finally, the resin was coated on the polished PVC plate and dried by ventilation. The PVC plate was hung at the Qingdao Port Wharf with an immersion depth of 1 m. The samples were salvaged regularly, the surface sediment was washed with seawater, and the number of fouling organisms was photographed and recorded.

3. Results and discussion

3.1. Fluorescence intensity analysis of carbazole derivatives

As shown in Fig. 3, PCAM has the highest fluorescence intensity because of the highest degree of conjugation. The fluorescence intensity

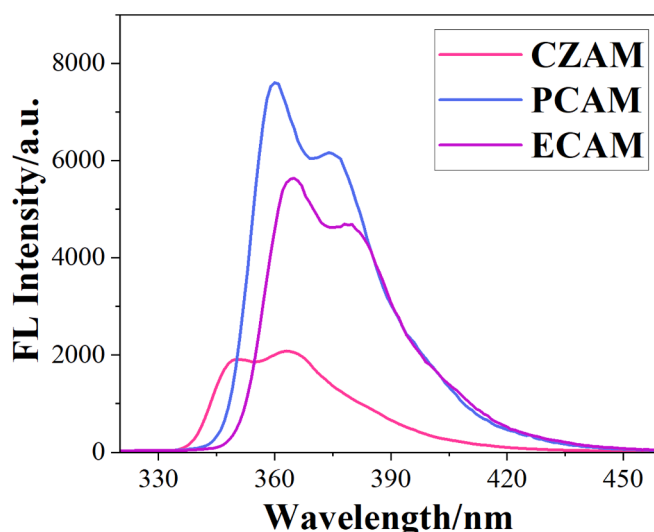


Fig. 3. Fluorescence emission spectra of carbazole derivatives.

of ECAM takes second place, and CZAM is the weakest, but its fluorescence intensity still exceeds 2000. Therefore, all three compounds have good fluorescence.

3.2. Analysis of bacteriostatic results of carbazole derivatives

Fig. 4 illustrates the inhibitory effect of carbazole derivatives on marine bacteria, with CN utilized as a control. A blank resulted in a substantial growth of colonies, whereas the fewest colonies were observed in the CN control. This occurred because CN, identified as the broad-spectrum fungicide chlorothalonil found on the market, possesses greater toxicity towards bacteria, thus achieving the highest antibacterial rate. Among the three experimental groups, PCAM exhibited an antibacterial rate of 90 %, which was comparable to that of CN. This stemmed from PCAM containing two amide groups and exhibiting high chemical activity, thereby exerting a pronounced inhibitory effect on the growth and reproduction of bacteria. CZAM demonstrated a bacteriostatic rate of 80 %, and ECAM displayed the lowest bacteriostatic rate; however, ECAM still exhibited a measurable bacteriostatic effect when compared to the blank. The experimental outcomes indicate that the three carbazole derivatives have a discernible inhibitory effect on marine *Pseudomonas*.

3.3. Analysis of algae inhibition test results of indole derivatives

The standard working curve of the algae solution is shown in Fig. S6 Supplementary Data. The concentration of algae solution showed a good linear relationship with absorbance.

The data in Fig. 5 show that CN has the most significant inhibitory effect on the growth of *Phaeodactylum tricornutum* and *Nitzschia closterium*, with an inhibition rate of 86 %. The growth inhibition rate of CZAM on *Phaeodactylum tricornutum* was the highest, reaching 81 %, and the inhibition effect was comparable to that of CN. The algae inhibition performance of PCAM is weaker than that of CZAM, which is because the solubility of PCAM is poor, resulting in a certain degree of influence on the algae inhibition rate. The inhibition rate of three carbazole derivatives on these two algae was more than 70 %, which showed that they had good algae inhibition performance. This inhibitory effect may be affected by the fluorescence and chemical activity of the compound, which significantly interferes with the normal physiological activities of microalgae through fluorescence exclusion, thus inhibiting the growth and reproduction of microalgae. According to the previous studies of the research group, it was found that indole derivatives containing amide groups could inhibit the formation of cell walls by promoting the outflow of Ca^{2+} in microalgae cells, thus inhibiting the growth and reproduction of microalgae. Since carbazole derivatives also contain similar amide groups, it can be inferred that these compounds may inhibit the growth and reproduction of microalgae through similar mechanisms, that is, inhibit the synthesis of microalgae cell walls.

3.4. Infrared spectra of zinc acrylate resins containing carbazole derivatives

As seen in Fig. 6, the telescopic vibrational absorption peak of $-\text{NH}-$ is at 3413 cm^{-1} , and the telescopic vibrational absorption peaks of $-\text{CH}_3$ and $-\text{CH}_2$ are at 2961 cm^{-1} and 2873 cm^{-1} . The carbonyl group in

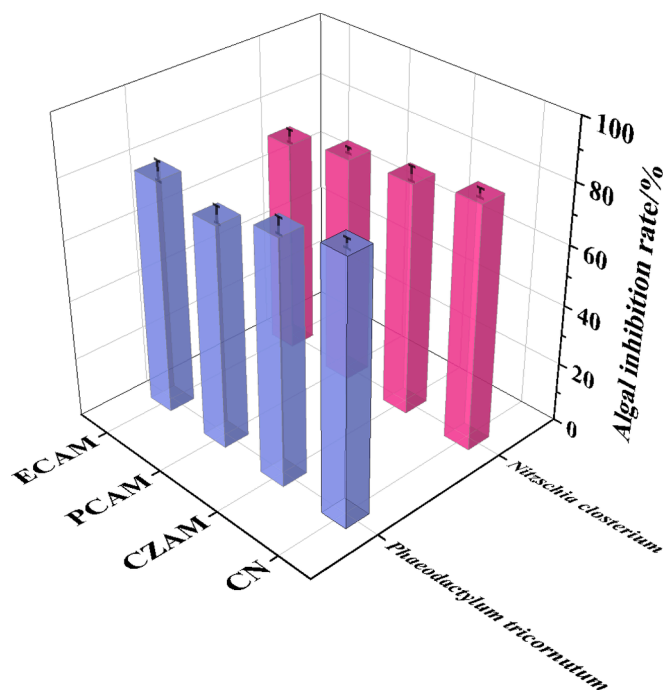


Fig. 5. Inhibition rate of carbazole derivatives on microalgae on the seventh day.

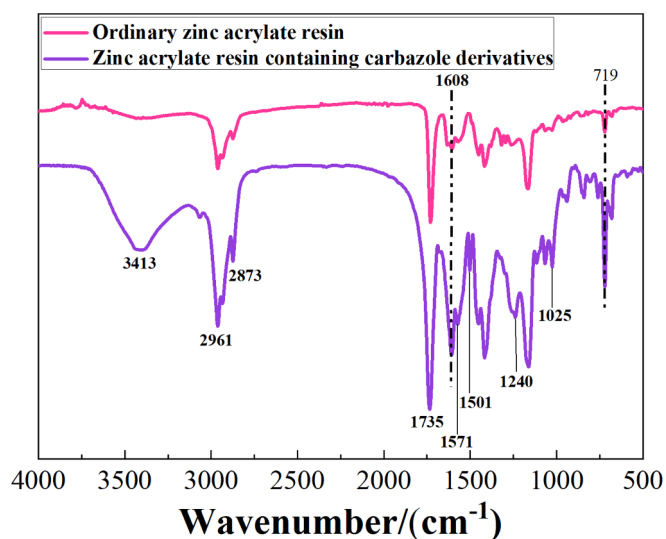


Fig. 6. IR comparison of common zinc acrylate resin with zinc acrylate resin containing carbazole derivative structures.

carboxylic acids has a telescopic vibrational absorption peak at 1735 cm^{-1} , and the telescopic vibrational absorption peak of $-\text{COO}-\text{Zn}-\text{COO}-$ is at 1608 cm^{-1} . 1571 cm^{-1} and 1501 cm^{-1} are the vibrational peaks of aromatic hydrocarbons, and the symmetric and asymmetric telescopic

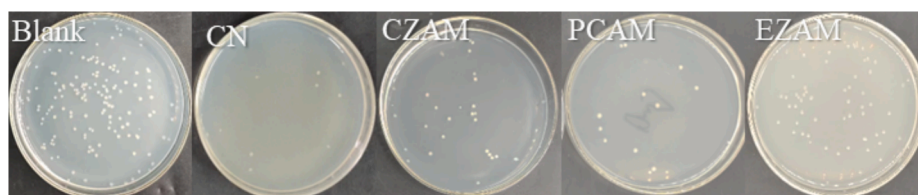


Fig. 4. Bacteriostasis diagram of carbazole derivatives.

vibrational absorption peaks of -COO- are at 1240 cm^{-1} and 1025 cm^{-1} , respectively. This indicates the introduction of carbazole derivatives in the zinc acrylate resin. At 719 cm^{-1} is the stretching vibration absorption peak of Zn-O . Thus, zinc acrylate resin containing carbazole derived structure was successfully synthesized.

3.5. Fluorescence intensity analysis of zinc acrylate resin containing carbazole derivative structures

The fluorescence emission spectra of the two resins are shown in Fig. 7. The resin that incorporated PCAM exhibited a higher fluorescence intensity compared to the one containing CZAM, which aligns with the inherent fluorescent properties of the respective compounds. An analysis was conducted on the fluorescence intensity variations of the two resins, revealing that their fluorescence intensity increased in correlation with the rising content of the carbazole derivative structure, demonstrating a robust regularity.

3.6. Analysis of algae inhibition results of zinc acrylate resin containing carbazole derivative structures

Fig. 8 shows the results of the algae inhibition performance of zinc acrylate resin containing carbazole derivative structure and ordinary zinc acrylate resin (control) on *Phaeodactylum tricornutum* and *Nitzschia closterium* under the two experimental conditions of sun light/dark alternating conditions and incubator light/dark alternating conditions. From the results, it can be seen that the algal inhibition rate of the experimental group was significantly higher than that of the control group under incubator light/dark alternating conditions, reaching a maximum of 88.5 %. This result clearly proves that the chemical repellency of the resin was significantly enhanced by the introduction of the fluorescent functional structure, and the algal inhibition effect increased with the increase of the content of fluorescent functional monomers.

The algal inhibition rate under sun light/dark alternating conditions was slightly lower than that under incubator light/dark alternating conditions. This may be since algae tend to photosynthesize more under natural light conditions and thus have a higher reproduction rate, while the chemical activity of the compounds did not change, resulting in a relatively lower algal inhibition rate under natural light conditions.

Comparing the different resins, the algal inhibition rate of CZ resin was higher than that of PCZ resin. This is because the algal inhibition effect of the resin is influenced by both the fluorescence and chemical activity of the compound. Although the fluorescence intensity of CZ resin is not as strong as that of PCZ, the molecular weight of CZAM in CZ resin is smaller, the permeability is higher, and it is easier to enter the interior of algal cells and destroy organelles, so its inhibition rate is higher than that of PCZ resin.

In summary, zinc acrylate resin containing carbazole derived structures is an efficient antifouling material that can effectively inhibit the growth of algae and has potential applications.

3.7. Analysis of anti-algae adhesion effect of zinc acrylate resin containing carbazole derivative structure

The algae attached to the resin surface were rinsed thoroughly, and chlorophyll (a) was detected using a portable fluorometer, and the results are shown in Fig. 9.

As can be seen from Fig. 9(A), under the alternating conditions of incubator light/dark, the amount of *Nitzschia closterium* attachment on the surface of the resin of the experimental group was significantly smaller than that of the control group, and the anti-adhesion rate could be maximally enhanced by 83.52 %. From Fig. 9(B), it can be seen that the anti-adhesion rate of the resin of the experimental group can be enhanced by 74.18 % compared with that of the control group, which proves that the chemical repellent anti-algal sedimentation and adhesion performance of the resin is significant after the introduction of carbazole derivatives. The amount of chlorophyll (a) on the surface of the resin was significantly lower than that in the incubator light/dark alternating condition under sun light/dark alternating condition, and the decrease of chlorophyll (a) increased with the increase of fluorescent functional monomer content, and the inhibition of attachment rate could be maximally elevated by 91.43 % and 97.37 % for the *Phaeodactylum tricornutum* and *Nitzschia closterium*, respectively. PCZ-6 had the highest fluorescence intensity and the best anti-adhesion performance under alternating sun light/darkness conditions, with a maximum enhancement of 47.27 % in the anti-algal attachment rate of the resin compared with the incubator light/darkness conditions, which suggests that the fluorescence repellency effect of the resin, in addition to the chemical repellency effect, is also an important factor in the excellent anti-algal deposition adhesion performance of the resin.

3.8. Analysis of the effect of zinc acrylate resin containing carbazole derivative structure on algae photosynthetic pigments

As can be seen from Fig. 10, the content of photosynthetic pigments in the *Phaeodactylum tricornutum* was slightly higher in the alternating sun light/darkness condition than in the alternating incubator light/darkness condition. This difference may be attributed to the fact that stronger fluorescence was emitted by the resin under natural light conditions, which could have facilitated photosynthesis in the algae, resulting in a smaller reduction of photosynthetic pigments.

Further analysis showed that after the introduction of carbazole derivative structure, the contents of chlorophyll (a), chlorophyll (c), and carotenoids in *Phaeodactylum tricornutum* in the experimental group were significantly reduced, whether in the incubator light or the

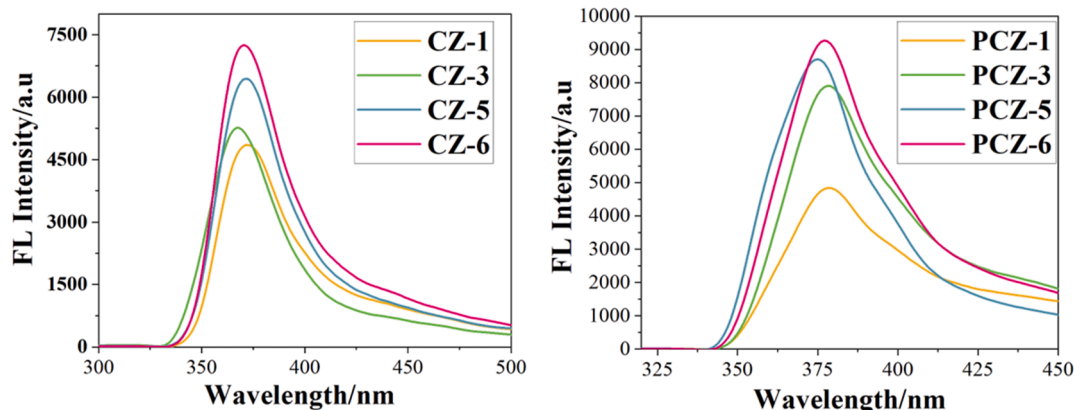


Fig. 7. Fluorescence intensity of zinc acrylate resin containing carbazole derivative structures.

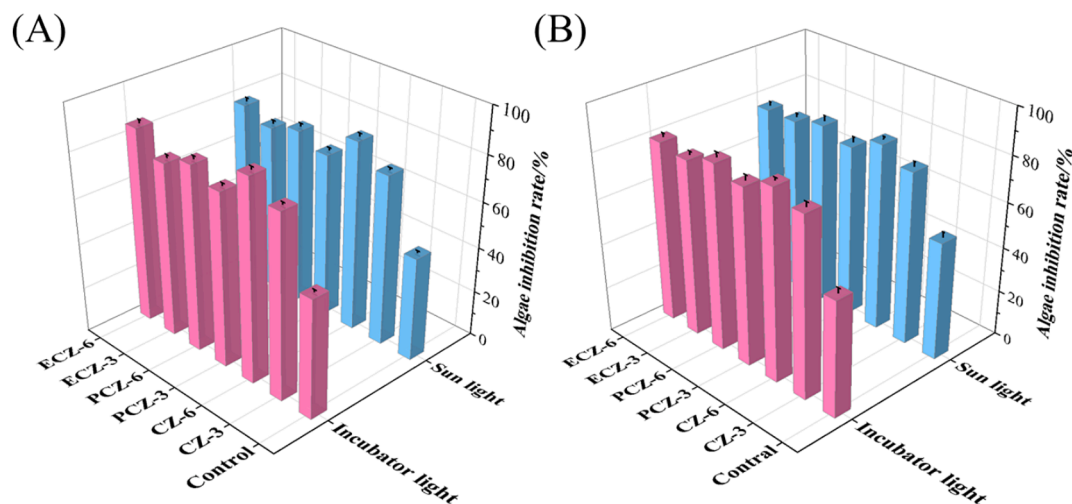


Fig. 8. Inhibition rate of coating on microalgae: A. *Phaeodactylum tricornutum* B. *Nitzschia Closterium*.

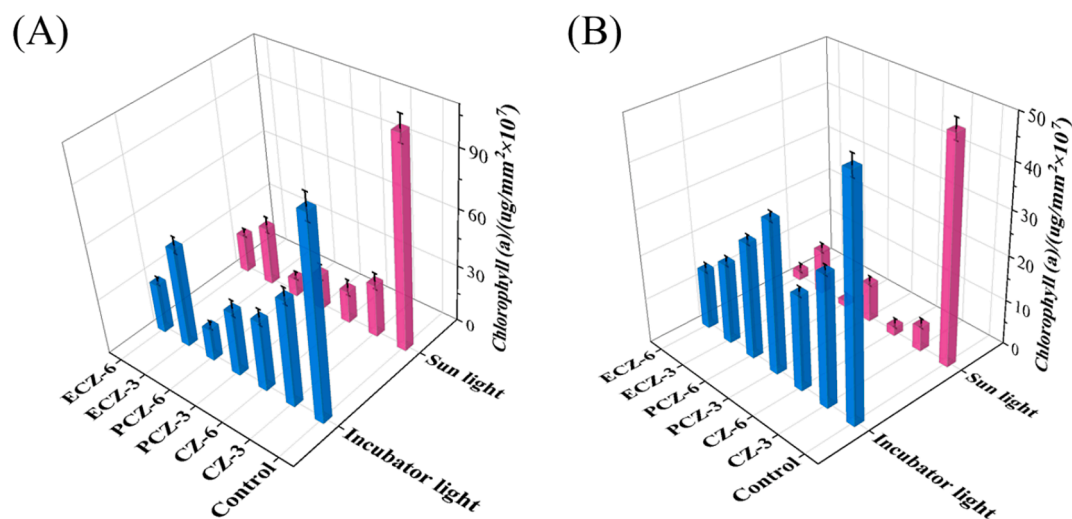


Fig. 9. Chlorophyll (a) content of algae deposited on the coating surface: A. *Phaeodactylum tricornutum* B. *Nitzschia Closterium*.

sunlight/dark conditions. Among the carotenoids, the resin CZ had the most significant effect on carotenoids, which were reduced by 80.77 % and 77.34 % under alternating incubator light/dark conditions and under alternating sun light/dark conditions, respectively. The effect of resin ECZ on chlorophyll (a) content was also considerable under alternating conditions of incubator light/darkness with a reduction of 49.46 % compared to the control.

The results of the algal inhibition experiment and the photosynthetic pigment content influence experiment under the incubator light condition showed that the carbazole-derived structure in the resin had good chemical activity and could effectively inhibit the growth and attachment of algae. The algal inhibition performance of the resin under solar light conditions was weaker than that of the incubator light conditions, while the anti-algal deposition performance was significantly better than that of the incubator, which preliminarily proved that the strong fluorescence property of the carbazole-derived structure made the algae inclined to photosynthesize rather than selectively settle, and, therefore, it was beneficial to prevent algal fouling.

3.9. Effect of zinc acrylate resin containing carbazole derivative structures on the morphology of algal cells

Fig. 11 shows the cell surface changes before and after the inhibition

experiment of zinc acrylate resin containing carbazole derivative structure on the growth of *Phaeodactylum tricornutum*. It can be observed that before the experiment, the cells of *Phaeodactylum tricornutum* in the control group A showed the typical shuttle-shaped morphology, and the cells were full and the cell wall was complete. After the experimental treatment, the algal cells in experimental group B showed wrinkled morphology, and cell membrane damage, and some cells even showed serious breakage and death. The results suggest that the amide group in carbazole derivatives may lead to Ca^{2+} efflux from the algal cells, which in turn damages the cell wall and is the main cause of cell damage.

3.10. Analysis of self-polishing performance of zinc acrylate resin containing carbazole amide compounds

By observing the microscopic morphology of the resins before and after the dynamic acceleration experiments by laser scanning confocal microscope, we obtained the experimental results in Table 1. From the results, it was seen that all the resin surfaces were relatively smooth and free of cracking before the dynamic acceleration experiments. After five cycles of experimentation, the resin surface of the control group showed a large number of scratches and was accompanied by cracking. In contrast, only a few scratches appeared on the surface of the resin in the experimental group, and no cracking was observed. This result clearly

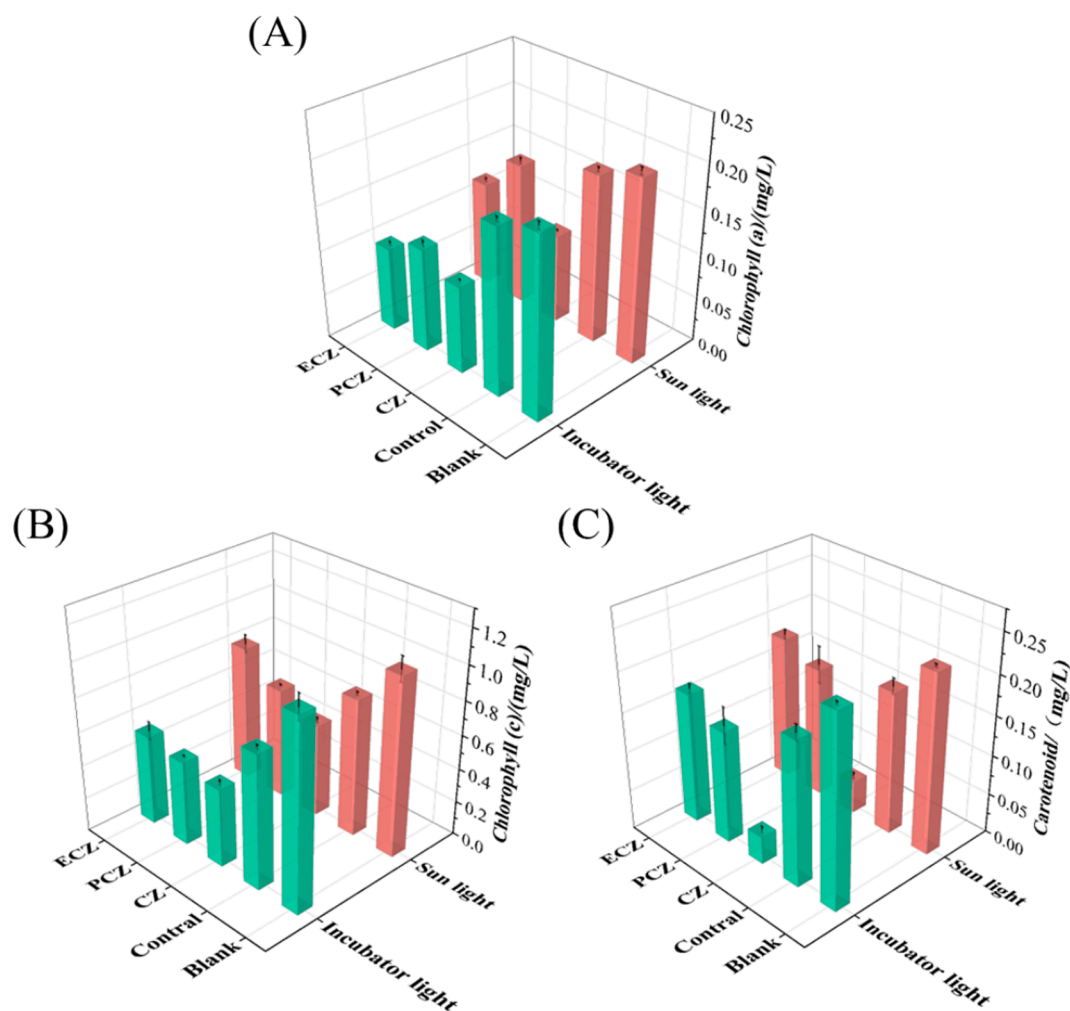


Fig. 10. Photosynthetic pigment content in *Phaeodactylum tricornutum*: A. Chlorophyll (a) B. Chlorophyll (c) C. Carotenoids.

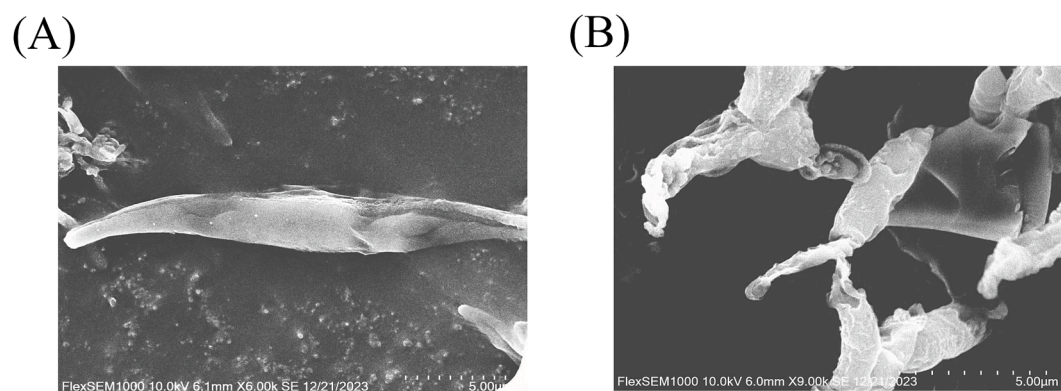


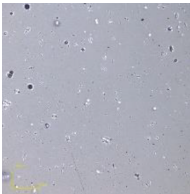
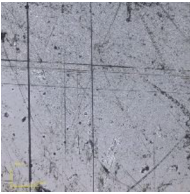
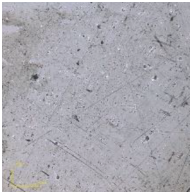
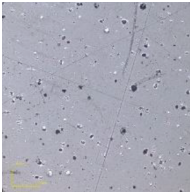
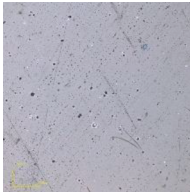
Fig. 11. Surface morphology of algal cells after algae inhibition test.

demonstrates that the introduction of carbazole amides effectively improves the surface properties of the resin and enhances its resistance to scratching and cracking.

Fig. 12 shows the mass loss of the resin in dynamic paddling. the abrasion rate of the resin was greater after the experiment was conducted for thirty days. there are three reasons for this, firstly, the solvent and small molecular weight monomers remaining in the resin are released, secondly, the hydrolysis of hydrophilic groups and the polishing effect on the resin surface, and finally, the ion exchange between

Na^+ in the seawater and Zn^{2+} in the resin. The abrasion rate of the resins stabilized as the experiment progressed. It is noteworthy that the abrasion rates of resins CZ, PCZ, and ECZ are significantly lower than those of the control group, indicating that the introduction of carbazole amide compounds improves the stability of the resins and reduces the abrasion rate of the resins, which can prolong the antifouling period effect of the resins.

Table 1
Surface morphology of the coating before and after 5 cycles of dynamic acceleration test.

	Control	CZ	PCZ	ECZ
Initial coating				
surface morphology for 5 cycles				

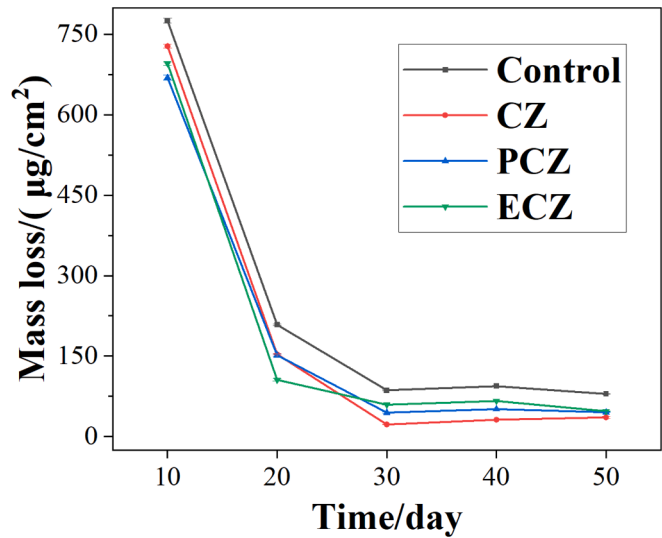


Fig. 12. Variation of mass abrasion rate of coating.

3.11. Marine antifouling performance test results of zinc acrylate resin containing carbazole derivative structures

As shown in Fig. 13, after 60 days of real sea anti fouling performance test, a large amount of biological fouling such as algae appeared on the surface of the control group, indicating that the biological fouling was quite serious. In the experimental group, with the increase of the

structural content of carbazole derivatives, the adhesion of fouling organisms decreased significantly, especially when the structural content of carbazole derivatives was 6 %. After 120 days of real-sea antifouling test, the coating still maintained excellent antifouling performance, and the coating surface with a carbazole derivative structure content of 6 % had only a small amount of fouling organisms and sludge. The experimental results show that the introduction of carbazole derivative structure can significantly improve the marine antifouling performance of zinc acrylate resin.

3.12. Antifouling mechanism of the coating

The antifouling mechanism of the resin is shown in Fig. 14. Firstly, in the anti-algae settling attachment experiment, the anti-algae attachment rate under natural light condition was 47.27 % higher than that under incubator light condition. It indicates that the fluorescent functional structure inhibits sinking attachment by interfering with the spatial structure of chloroplasts of algal cells, which makes the algae more inclined to the action of light, thus playing the role of fluorescence repellent [12,13].

In addition, in the algae inhibition experiment of resin, the inhibition rate of resin on algae growth reached 88.5 %. Combined with the scanning electron microscope images of algae cell surface before and after the algae inhibition experiment, it was proved that the zinc acrylate resin containing carbazole derivative structure could promote the outflow of Ca^{2+} in algae cells and inhibit the synthesis of cell wall. In the experiment of the effect of the resin on the photosynthetic pigment of algae, the content of carotenoids in algae cells decreased by 80.77 %, which proved that the resin inhibited the synthesis of photosynthetic pigments in algae cells. Therefore, the resin plays a chemical avoidance

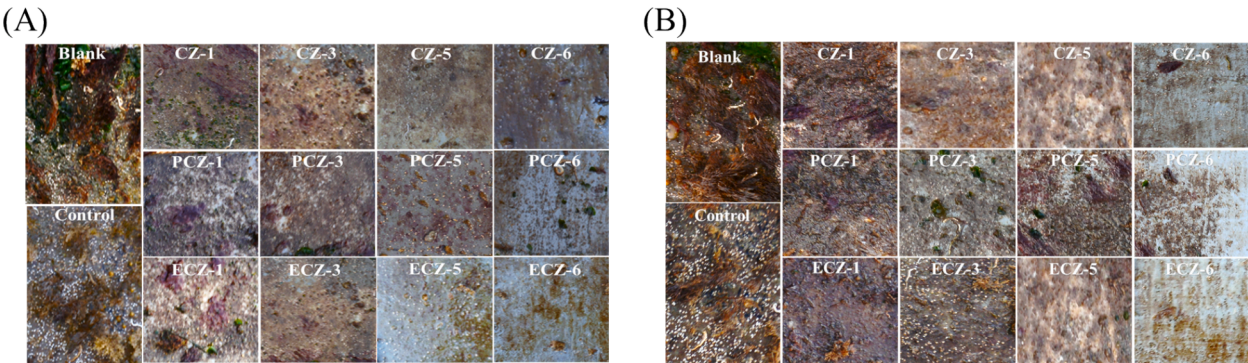


Fig. 13. Coated offshore hanging plate picture: A. 60 days B. 120 days.

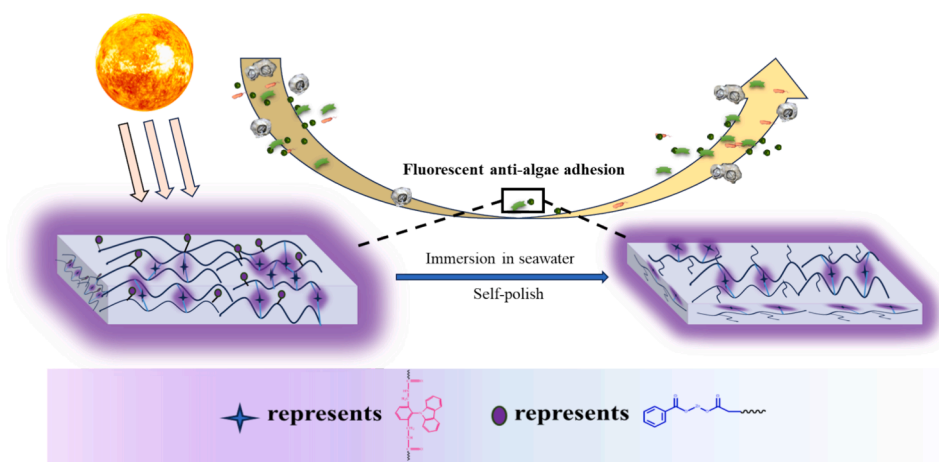


Fig. 14. Scheme of antifouling mechanism of coating.

effect on algae [23].

Finally, in the self-polishing experiment, the resin can exchange the metal group in the side chain with Na^+ and Mg^{2+} in the seawater and dissolve the hydrophilic group to realize the continuous update of the coating surface under constant water flow, which makes the attached fouling organisms peel off from the resin layer by layer under the erosion of seawater, effectively reducing the adhesion of fouling organisms [22,32,33].

In summary, the zinc acrylate resin containing carbazole derivative structure combines high chemical repellent performance, high fluorescence repellent performance and self-polishing performance, showing excellent antifouling performance, which provides a multi-purpose solution for the prevention and control of marine biofouling.

4. Conclusion

In this paper, three carbazole derivatives with fluorescent properties were successfully synthesized, and the three compounds exhibited significant growth inhibition effects on marine bacteria and microalgae. The zinc acrylate resin containing the above fluorescent functional structures exerted its algal inhibition by destroying the structure of microalgal cell membranes and interfering with the synthesis of cell walls and showed excellent performance in inhibiting algal growth, among which, the CZ-6 resin showed a high inhibition rate of 88.5 % against *Phaeodactylum tricornutum*, which increased the algal inhibition rate by 68.9 % compared with that of the blank group. The fluorescence property could significantly enhance the anti-algal adhesion rate of the resin, and the PCZ-6 resin could increase the anti *Nitzschia closterium* adhesion rate by 97.37 %. In addition, the introduction of carbazole-derived structures improved the stability of the resin and led to a significant increase in antifouling performance in the real sea.

CRediT authorship contribution statement

Wenjian Dong: Writing – original draft, Data curation. **Yifan Wang:** Data curation. **Xia Li:** Funding acquisition, Formal analysis. **Liangmin Yu:** Project administration, Methodology. **Na Li:** Supervision. **Jian Yang:** Investigation. **Junchao Yin:** Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by National Natural Science Foundation of China [grant number U22A20112], and the Hainan Provincial Joint Project of Sanya Yazhou Bay Science and Technology City [grant numbers 2021JJLH0013].

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cej.2024.158017>.

Data availability

Data will be made available on request.

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